Kemeny's Constant for Nonbacktracking Random Walks

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Nonbacktracking Random Walks

Definition 1. *A discrete, time-homogeneous* ***Markov chain*** *is a stochastic process on a finite state space where transitions depend only on the current state.*

Definition 2. *A* ***nonbacktracking random walk (NBRW)*** *disallows immediate return to the previous state. Thus it is not a Markov Chain on the vertices of a graph. On directed edges, it can be modeled as a Markov chain.*

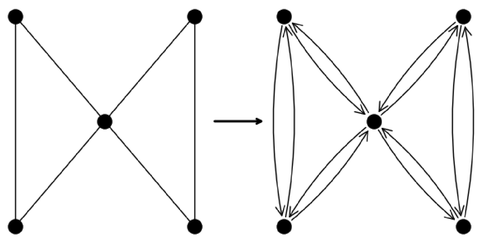

Figure 1: Walks on vertices of undirected graphs can be thought of as walks on the edges of directed graphs.

Kemeny's Constant

Definition 3. *The* ***stationary vector****, π, is the unique normalized and nonnegative vector such that*

$$\pi^T = \pi^T P$$

where P is the transition matrix of the Markov Chain. The Perron-Frobenius theorem guarantees the existence of π when P is irreducible and aperiodic.

Definition 4. *The* ***mean first passage time*** *m_{ij} is the expected time to reach state s_j starting from state s_i:*

$$m_{ij} := \mathbb{E}[F_{ij}],$$

where F_{ij} is the first time s_j is reached from s_i. The matrix of these values is called the ***mean first passage matrix****, $M = [m_{ij}]$.*

Theorem 1. ***Fundamental Matrix*** *(proved in [Kem81]):*
A generalized inverse of the matrix $I - P$ is given by

$$Z = (I - P + \mathbf{1}\pi^T)^{-1}.$$

Note that the mean first passage times satisfy the recursive relation

$$m_{ij} = \sum_{k \neq j} p_{ik} m_{kj} + 1 \qquad M = P(M - M_{dg}) + J$$

where M_{dg} is the diagonal matrix of the diagonal entries of M and J is the all ones matrix.
The fundamental matrix can be used to write

$$M = (I - Z + JZ_{dg})W^{-1} \qquad \text{where } W = \text{Diag}(\pi)$$

Definition 5. ***Kemeny's constant*** *$\mathcal{K}$ is defined as:*

$$\mathcal{K} := \sum_{j \neq i} m_{ij}\pi_j.$$

Remarkably, $\mathcal{K}$ is independent of the choice of state i. This quantity reflects how efficiently a Markov Chain explores the state space. It can also be shown that Kemeny's constant can be equivalently expressed as $\mathcal{K} = \text{tr}(Z) - 1$.

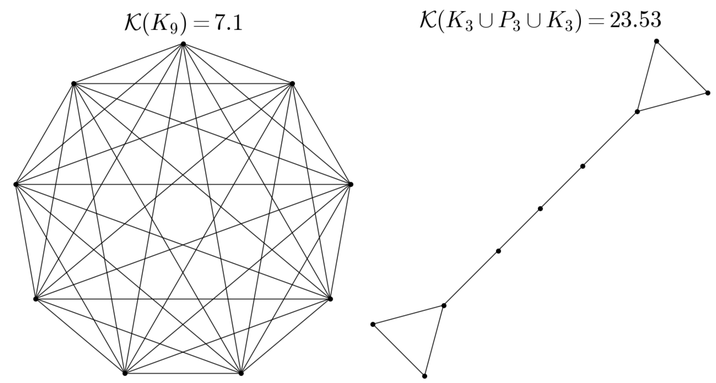

Figure 2: Extremal Graphs of Kemeny's constant for $|V(G)| = 9$.

Kemeny's Constant for NBRW

Theorem 2. ***Edge Space Kemeny's Constant*** *([BFG+23]):*
For a connected graph G with $|V(G)| = n$ and $|E(G)| = m$,

$$\mathcal{K}_e(G) = \mathcal{K}_v(G) + 2m - n.$$

Theorem 3. ***Nonbacktracking Edge Space Kemeny's Constant****:*
For a connected d-regular graph G ($d \geq 3$),

$$\mathcal{K}^e_{nb}(G) = \frac{(d-2)\mathcal{K}_e(G)}{d} + 2n + \frac{1}{d-2} - \frac{n}{d}.$$

Theorem 4. *For a d-regular graph G ($d \geq 3$), not K_4, K_5, or $K_{3,3}$,*

$$\mathcal{K}^e_{nb}(G) < \mathcal{K}_e(G).$$

Idea 1. *Extending Kemeny's constant for NBRW on vertices normalizes comparisons for graphs with the same number of vertices.* ***How to define this?***

Approach 1. ***Using MFPTs:***
For SRW, $\mathcal{K} = \pi^T M \pi$. For NBRW, let M^v_{nb} be the nonbacktracking mean first passage matrix. Define:

$$\mathcal{K}^v_{nb}(G) := \pi^T M^v_{nb} \pi.$$

Approach 2. ***Projection of Edge Space Fundamental Matrix:***
For NBRW, let $Z^e_{nb} = (I - P_{nb} + W_{nb})^{-1}$, where $W_{nb} = \mathbf{1}\pi_e^T$. Define the vertex space projection:

$$Z^v_{nb} = D^{-1}T(I - P_{nb} + W_{nb})^{-1}T^T, \qquad \text{allowing } \mathcal{K}^v_{nb}(G) := \text{tr}(Z^v_{nb}) - 1.$$

Theorem 5. *For connected edge-transitive graphs G,*

$$\pi^T M^v_{nb} \pi = \text{tr}(Z^v_{nb}) - 1.$$

Approach 3. *As $\mathcal{K}_e(G) = \mathcal{K}_v(G) + 2m - n$ holds for SRW, a similar result may hold for NBRW:*

$$\mathcal{K}^v_{nb}(G) = \mathcal{K}^e_{nb}(G) + 2m - n.$$

Empirically, discrepancies are within an order of magnitude.

Conjecture 1. *For all G,*

$$\text{tr}(Z^v_{nb}) - 1 \leq \pi^T M^v_{nb} \pi \leq \mathcal{K}^e_{nb} + 2m - n.$$

Conjecture 2. *For any connected graph G,*

$$\mathcal{K}^v_{nb}(G) < \mathcal{K}_v(G).$$

Conjectured Extremal Graphs

Conjecture 3. *The graph on n vertices that minimizes $K^v_{nb}(G)$ is the graph C_nC_3, an example of which is shown below.*

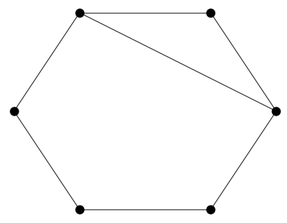

Figure 3: The graph C_6C_3 is a cycle on 6 vertices that contains an embedded cycle of length 3.

Conjecture 4. *The graph on n vertices that maximizes $K^v_{nb}(G)$ is the barbell graph with the shortest path length connecting two evenly sized complete graphs.*

References

[BFG+23] Jane Breen, Nolan Faught, Cory Glover, Mark Kempton, Adam Knudson, and Alice Oveson, *Kemeny's constant for nonbacktracking random walks*, Random Structures & Algorithms **63** (2023), no. 2, 343–363.

[Kem81] John G Kemeny, *Generalization of a fundamental matrix*, Linear Algebra and its Applications **38** (1981), 193–206.